# Item Response Theory

Course Syllabus

Instructor: Hau-Hung Yang

## Course Information

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| **Course Level:** | Graduate |
| **Credits:** | 3 credits |
| **Prerequisites:** | Graduate-level statistics (regression, maximum likelihood estimation); basic familiarity with R |
| **Software:** | R (packages: `mirt`, `ltm`, `catR`, `TAM`) |

## Course Description

This graduate-level course provides a comprehensive introduction to item response theory (IRT), the dominant psychometric framework for modeling the relationship between latent traits and observed responses to test items. The course covers the theoretical foundations of major IRT models, parameter estimation methods, model fit evaluation, and practical applications including test construction, differential item functioning (DIF) analysis, and computerized adaptive testing (CAT). Emphasis is placed on understanding the mathematical underpinnings of IRT models, developing the ability to conduct IRT analyses with real data, and critically evaluating the assumptions and limitations of IRT in applied settings.

## Course Objectives

Upon completion of this course, students will be able to:

1. Describe the assumptions and mathematical forms of major IRT models: Rasch/1PL, 2PL, 3PL, graded response, and generalized partial credit.
2. Understand and apply parameter estimation methods, including marginal maximum likelihood (MML), conditional maximum likelihood (CML), and Bayesian estimation (EAP, MAP).
3. Evaluate model fit and conduct model comparison using appropriate indices and graphical diagnostics.
4. Perform DIF analysis to assess measurement fairness across groups.

5. Understand the principles of computerized adaptive testing and implement basic CAT simulations.
6. Apply IRT methods to real datasets using R and interpret results in substantive contexts.

## Textbooks and References

- de Ayala, R. J. (2022). *The Theory and Practice of Item Response Theory* (2nd ed.). Guilford Press. **[Primary textbook]**
- Baker, F. B., & Kim, S.-H. (2017). *The Basics of Item Response Theory Using R*. Springer.
- van der Linden, W. J. (Ed.) (2016). *Handbook of Item Response Theory* (Vols. 1–3). CRC Press.
- Supplementary journal articles (distributed by topic); machine-learning applications are optional extension readings.

## Assessment

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| Homework assignments (theory + R exercises) | 30% |
| Midterm examination | 20% |
| Final examination | 20% |
| Applied IRT project (written report + presentation) | 20% |
| Paper reading summaries and class discussion | 10% |

## Weekly Schedule

| **Week** | Topic |
|---|---|
| **1** | Introduction: From classical test theory to IRT; limitations of CTT; overview of latent trait models |
| **2** | The Rasch model (1PL): Model formulation, sufficient statistics, specific objectivity, and the conditional maximum likelihood estimator |
| **3** | The two- and three-parameter logistic models (2PL, 3PL): Item discrimination, difficulty, and guessing; item characteristic curves (ICCs) and item information functions; identification issues |
| **4** | Parameter estimation I: Joint, conditional, and marginal maximum likelihood estimation; the EM algorithm |

| **Week** | Topic (continued) |
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| **5** | Parameter estimation II: Bayesian estimation methods (EAP, MAP); prior specification and its impact |
| **6** | Ability estimation: Maximum likelihood, Bayesian, and weighted likelihood estimators; standard errors and confidence intervals |
| **7** | Test and item information: Fisher information, test information function, and its role in test design |
| **8** | *Midterm Examination* |
| **9** | Model fit and comparison: Likelihood-ratio tests, AIC/BIC, and graphical diagnostics; overview of multidimensional IRT and its relation to factor analysis |
| **10** | Models for polytomous data: The graded response model (GRM), the (generalized) partial credit model, and the nominal response model |
| **11** | Differential item functioning (DIF): Concepts, IRT-based detection methods (Lord's $\chi^2$, likelihood-ratio test, Raju's area measure) |
| **12** | Test construction and item bank development: Item selection criteria, information-based design, and equating |
| **13** | Computerized adaptive testing I: Principles, maximum-information and Bayesian item selection, stopping rules; simulation in R |
| **14** | Computerized adaptive testing II: Exposure control and content balancing; online-calibration overview; project feedback and AI-analysis critique |
| **15** | Applied IRT project presentations and discussion |
| **16** | *Final Examination* |

The schedule follows the 16-week NTU semester: fourteen teaching weeks, the midterm examination in Week 8, and the final examination in Week 16. A class lost to a national holiday is made up in the flexible week after Week 16 (Week 16+1), as the NTU academic calendar allows.

## Applied IRT Project

Each student (or group of two) will complete an applied IRT project involving a real or publicly available dataset. The project report (approximately 10–15 pages) should include:

1. A description of the dataset, research context, and the purpose of the analysis.
2. Preliminary item analysis using classical test theory indices.
3. Model fitting: Fit at least two IRT models (e.g., Rasch vs. 2PL, or GRM vs. GPCM) and compare them using appropriate fit indices.
4. Parameter estimation results with interpretation of item and ability parameters.
5. At least one applied analysis, such as DIF detection, test information evaluation, or a CAT simulation based on the fitted item bank.
6. A discussion of the findings, limitations, and implications for test development or revision.

In Week 14, students will compare an AI-generated model interpretation with their own analysis, check model assumptions and parameter interpretations, and document any corrections. The instructor can provide the AI output. Include this brief critique in the project report; it is assessed as part of the project grade. Reports, reproducible R code, and presentations are due in Week 15.

## Course Policies

- All homework and project analyses must be submitted as reproducible R scripts or R Markdown documents.
- Late submissions will incur a 10% penalty per day, up to a maximum of three days.
- Students are expected to read assigned papers before class and participate actively in discussions.
- AI assistance must be disclosed and checked independently. Use public or simulated data for external AI tools; instructor-provided outputs can be used for the critique.